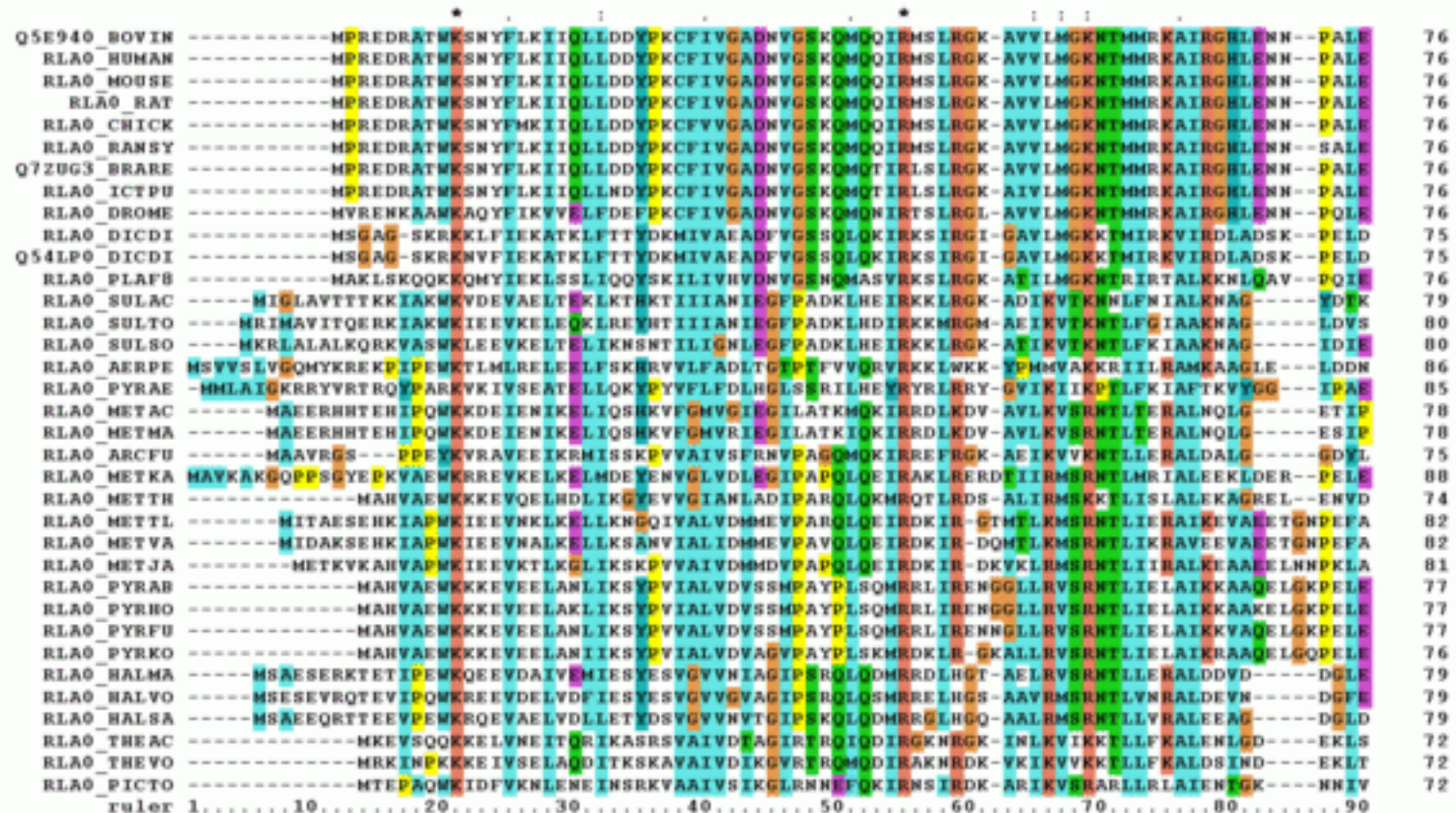
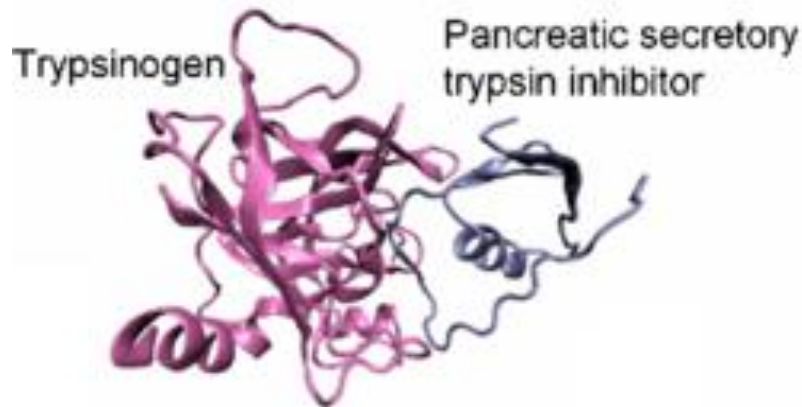


- a quick overview -



Motivations for sequence alignment

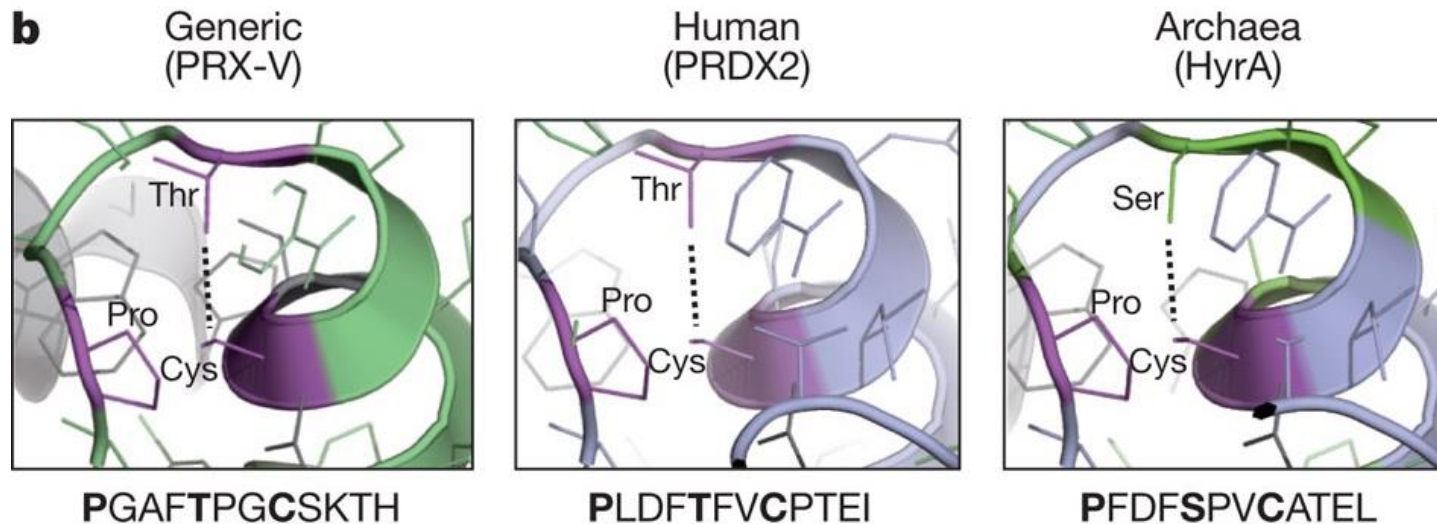
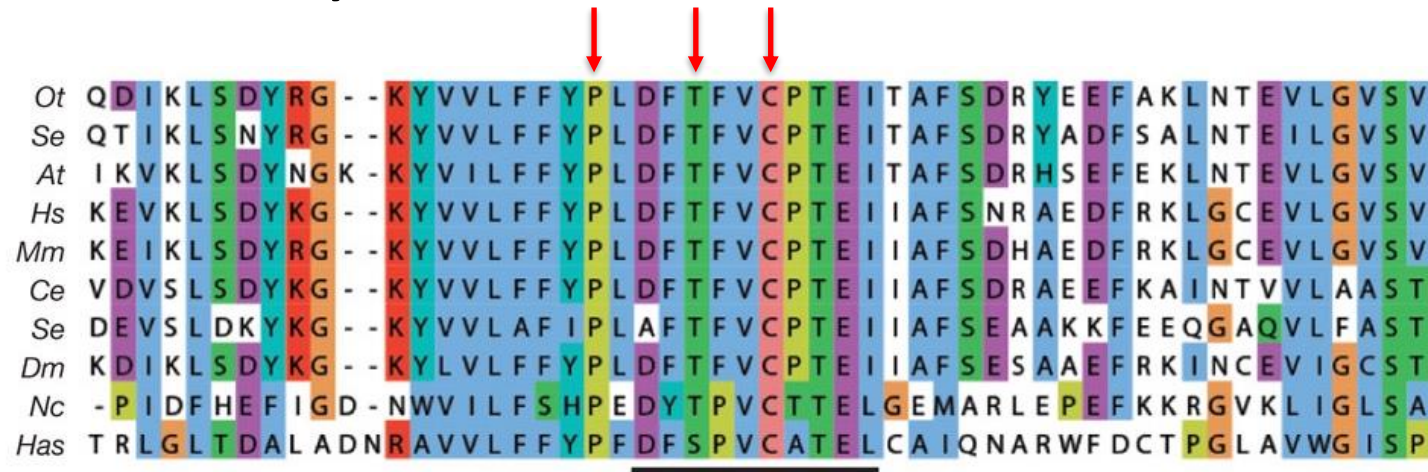
1) find genes that are related by common descent



Chymotrypsin	VKKTMCAGG-DGVISACNGDSGGPLNCQLENGSWEVFGIVSFGSRRGC [...]
	+ M C G +G +C GDSGGP+ C NG + G+VS+G GC
Trypsinogen	ITSNMFCVGFLEGGKDSCQGDGGPVVC---NGQLQ--GVVSWGD--GC [...]

Motivations for sequence alignment

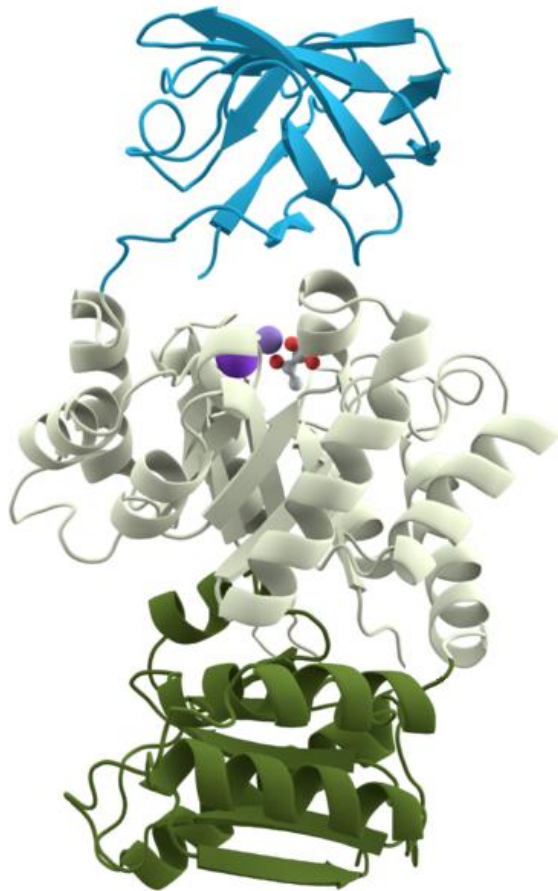
2) to identify and check the state of “active sites”



From: Peroxiredoxins are conserved markers of circadian rhythms. Nature 485, 459–464 (24 May 2012)

Motivations for sequence alignment

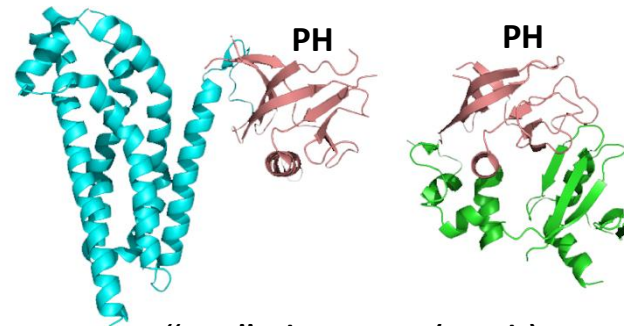
3) to identify and characterize “protein domains”



Pyruvate Kinase

definition:

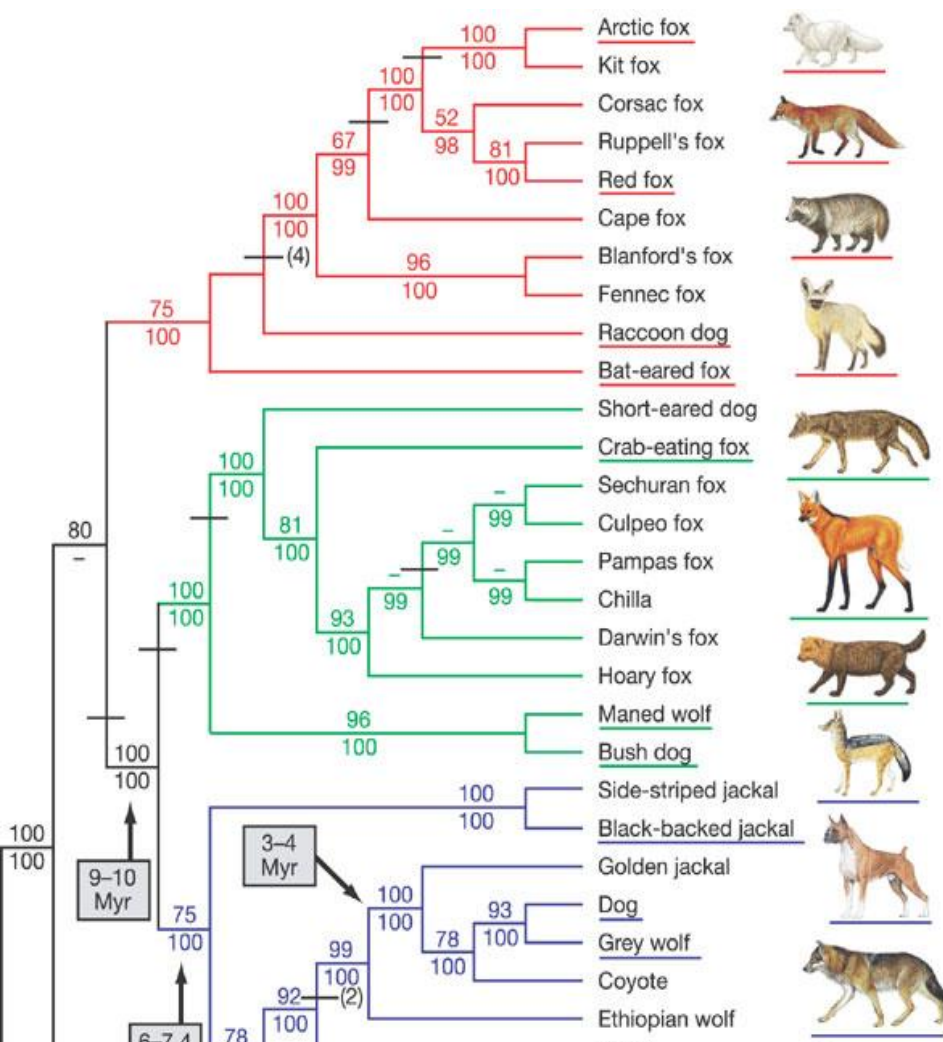
“parts of proteins that can evolve, function, and exist independently of the rest”



“PH”-domain (pink);
occurring in two different proteins

Motivations for sequence alignment

4) to make phylogenetic inferences (“trees”)



canine 1

canine 2

canine 3

[...]

IRMSLRGK	-	AVVLMGKNTMMRKAIRGHLENN	--	PALE
IRMSLRGK	-	AVVLMGKNTMMRKAIRGHLENN	--	PALE
IRMSLRGK	-	AVVLMGKNTMMRKAIRGHLENN	--	PALE
IRMSLRGK	-	AVVLMGKNTMMRKAIRGHLENN	--	PALE
IRMSLRGK	-	AVVLMGKNTMMRKAIRGHLENN	--	PALE
IRMSLRGK	-	AVVLMGKNTMMRKAIRGHLENN	--	SALE
IRLSLRGK	-	AVVLMGKNTMMRKAIRGHLENN	--	PALE
IRLSLRGK	-	AIVLMGKNTMMRKAIRGHLENN	--	PALE
IRTSLRGL	-	AVVLMGKNTMMRKAIRGHLENN	--	PQLE
IRKSIRGI	-	GAVLMGKKTMIKRVIRDLADSK	--	PELD
IRKSIRGI	-	GAVLMGKKTMIKRVIRDLADSK	--	PELD
VRKSLRGK	-	ATILMGKNTIRIATLKKNLQAV	--	PQIE
IRKKLRGK	-	ADIKVTKNLNFNIALKNAG	----	YDTK
IRKKMRGM	-	AEIKVTKNTLFGIAAKNAG	----	LDVS
IRKKLRGK	-	ATIKVTKNTLFIKIAAKNAG	----	IDIE
VRKKLWKK	-	YPMVAKKRIILRAMKAAGLE	----	LDDN
YRYRLRY	-	GVIKIIPKPTLFIKIAFTKVYGG	----	IPAE
IRRDLDKV	-	AVLKVSNTLTERALNQLG	----	ETIP
IRRDLDKV	-	AVLKVSNTLTERALNQLG	----	ESIP
IRREFRGK	-	AEIKVYKNTLLERLALDAG	----	GDYL
IRAKLRERD	-	TIIRMSRNTLMRIALEEKLDER	--	PELE
MRQTLRDS	-	ALIRMSKKTLLISLALEKAGREL	--	ENVD
IRDKIR	-	GTMTLKMSRNTLIERAIKEVAEETGNPEFA	--	ENVD
IRDKIR	-	DQMTLKMSRNTLIKRAVEEVAEETGNPEFA	--	ENVD
IRDKIR	-	DKVKLRMSRNTLIIRALKEAAEELNPKLA	--	ENVD
MRRLIRENG	-	GLLRVSRNTLIELAIKKAAGELGKPELE	--	ENVD
MRRLIRENG	-	GLLRVSRNTLIELAIKKAAGELGKPELE	--	ENVD
MRRLIRENG	-	GLLRVSRNTLIELAIKKAAGELGKPELE	--	ENVD
MRDKLR	-	GKALLRVSRNTLIELAIKKAAGELGKPELE	--	ENVD
MRDLHGT	-	AELRVSRNTLLEALDDVD	----	DGLE
MRRELHGS	-	AAVRMSRNTLVNRLDEVN	----	DGFE
MRRLHGO	-	AALRMSRNTLLVRALEEAG	----	DGLD
IRGKNRGK	-	INLKVIKKTLLFKALENLGD	----	EKLS

How it's done ...

Chymotrypsin VKKTMVCAGG-DGVISACNGDSGGPLNCQLENGSWEVFGIVSFGSRRGC [...]
 + M C G +G +C GDSGGP+ C NG + G+VS+G GC
 Trypsinogen ITSNMFCVGFLEGGKDSCQGDSSGGPVVC---NGQLQ--GVVSWGD--GC [...]

NCQLENGSWEV
 C NG +
 VC---NGQLQ-

=> “substitution matrix” (learned from structures)

- Why not align “QL” instead of “NG” ??
- What does the “+” mean ?
- How “good” is my alignment?

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
A	4	-1	-2	-2	0	-1	-1	0	-2	-1	-1	-1	-1	-2	-1	1	0	-3	-2	0
R	-1	5	0	-2	-3	1	0	-2	0	-3	-2	2	-1	-3	-2	-1	-1	-3	-2	-3
N	-2	0	6	1	-3	0	0	0	1	-3	-3	0	-2	-3	-2	1	0	-4	-2	-3
D	-2	-2	1	6	-3	0	2	-1	-1	-3	-4	-1	-3	-3	-1	0	-1	-4	-3	-3
C	0	-3	-3	-3	9	-3	-4	-3	-3	-1	-1	-3	-1	-2	-3	-1	-1	-2	-2	-1
Q	-1	1	0	0	-3	5	2	-2	0	-3	-2	1	0	-3	-1	0	-1	-2	-1	-2
E	-1	0	0	2	-4	2	5	-2	0	-3	-3	1	-2	-3	-1	0	-1	-3	-2	-2
G	0	-2	0	-1	-3	-2	-2	6	-2	-4	-4	-2	-3	-3	-2	0	-2	-2	-3	-3
H	-2	0	1	-1	-3	0	0	-2	8	-3	-3	-1	-2	-1	-2	-1	-2	-2	2	-3
I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4	2	-3	1	0	-3	-2	-1	-3	-1	3
L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4	-2	2	0	-3	-2	-1	-2	-1	1
K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5	-1	-3	-1	0	-1	-3	-2	-2
M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5	0	-2	-1	-1	-1	-1	1
F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6	-4	-2	-2	1	3	-1
P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7	-1	-1	-4	-3	-2
S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4	1	-3	-2	-2
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-2	-2	0
W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11	2	-3
Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	-1
V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4

Pairwise Alignment

a) BLAST: quick and dirty

VKKTMCAGG-DGVISACNGDSGGPLNCQLENGSWEVFGIVSF
GGP

1) identify word-matches
(use indexing)

← VKKTMCAGG-DGVISACNGDSGGPLNCQLENGSWEVFGIVSF →
+ M C G +G +C GDSGGP+ C NG + G+VS+
ITSNMFCVGFLEGGKDSCQGDSGGPVVC---NGQLQ--GVVSW

2) extend into an “HSP”
(high-scoring sequence pair)

b) Dynamic Programming – correct and slow

	A	T	T	G	G	A	G	T	C	G
A	1	⇒0	⇒-1	⇒-2	⇒-3	⇒-4	⇒-5	⇒-6	⇒-7	⇒-8
T	↓0	↘2	↖1	→0	→-1	→-2	↘-3	↖-4	↖-5	↖-6
C	↓-1	↓1	↘2	→1	→0	→-1	↘-2	↖-3	↖-3	↖-4
T	↓-2	↘0	↖2	→2	→1	→0	↘-1	↖-1	↖-2	↖-3
C	↓-3	↓-1	↘1	→2	→2	→1	↘0	↖-1	↖0	↖-1
G	↓-4	↓-2	↘0	→2	→3	→2	↘2	↖1	↖0	↖1

ATTGGAGTCG

83%

ATC-----TCG

OR

ATTGGAGTCG

83%

AT-----CTCG

explore all possible paths
(but excluding obvious
dead ends)

Multiple Alignment

```

      10      20      30      40      50      60      70      80      90      100      110      120      130
1 -----MALKEKSLVRLVRLVLI-----LVLGWVQPSLGKE-SRAKKFQRQHMDSDSSPSSSS-----TYCNQMRR--RNMTQ 65
1 -----MVP-KLFTSQICLLLLLGL-----LAVEGSLHVKPPQFTWAQWFETQHINMTSQ-----QCTNAMQV--INNYYQ 61
1 -----MVP-KLFTSQICLLLLLGL-----MVEGSLHARPPQFTRAQWFAIQHISLNPP-----RCTIAMRA--INNYYR 61
1 -----MALQRTHSLLLLLLLTL-----LGLGLVQPSYGQD-GMYQRFLRQHVHPEETGGSD-----RYCNLMMQR--RKMTL 64
1 -----MVMG-----LGVLLLVFV-----LGLGLTPPTLAQDNSRYTHFLTOHYDAKPPQGRDD-----RYCESIMRR--RGLTS 61
1 -----MVLCPF-LLLLLLVL-----WGPVCPHAWPKRLTKAHWFEIQHIQPSPL-----QCNRAMSG--INNYYT 57
1 -----MAPARAGFCPLLLLLLLGL-----WVAEIPVSAKPKGMTSSQWFKIQHMQPSQ-----ACNSAMKN--INKHT 62
1 -----MAPARAGCCP-LLLLLLGL-----WVAEVLVRAPPKDMTSSQWFKTQHVQPSQ-----ACNSAMS I--INKYT 61
1 MMRTLITTHPLPLL LLPQQLQLVQFQEVDTDFDFPEEDKKEEFEECLEKFFSTGPARPPTKEKVKRRVL IEPG-----MPLNHIEYCNHEIMG-KNVYYK 95
1 -----MKLNLVQIFFMLMLLLGLGMGLGLSLHMAATAVLEESDQPLNEFWSSDSQDKAEATEEGDGTQTTETLVLSNKEVVQPGWPEDPILGEDEVGGNKMLRASALFQSNKDYLRDQTDRECNDDMAHK-MKEPS 131
1 -----METFPLLLSLGLVLAEESESTMKI I KEEFTDEEMQYDMAKSGQEKQTIE I LMNP I LLVKN-----TSLSMKDDMSSTLLTFRSLHYNDPKGNSSGNDKECCNDMTVWRKVSEAN 111
1 -----MIIMV I IFLVLLF-----WENEVNDEAVMSTLEHLHVDYPQNDVFPVPAR-----YCNHMI I QRV I REPD 59
1 -----MAPAVTRLFLQLVLG--PTLVMDIKMQIGSRNFYTLSIDYPRVNYPKGFRG-----YCNGLMSYMRGKMQN 65
1 -----MALK-SLVLLSLLVLVL-----LLV-RVQPSLGKE-TAAAKFERQHMDSSTSAASS-----NYCNQMKS--RNLT 63

```

Combinatorial Explosion: very many possible solutions

Complexity: $O(\text{alignment_length}^{\text{number_seqs}})$

=> an NP-complete problem !!

Multiple Alignment

TCOFFEE

PROBCONS

MUSCLE



SATé



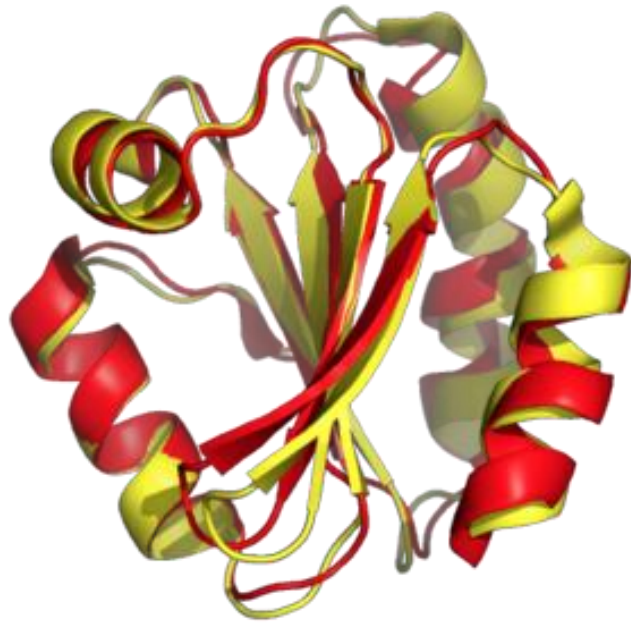
MAFFT



PRANK

DIALIGN-TX

Quality of MSA: Benchmarking



“BALiBASE”:

Benchmark Alignment Database

Hand-made multiple sequence alignments
Based on selected structural alignments

Structural Alignments
offer the best
benchmarks !

Welcome to BALiBASE 4

[download the whole benchmark by html](#)

Reference 1: variability, length

Reference 1: variability, length

Reference 2: orphans

Reference 3: sub-families

Reference 4: extensions

Reference 5: insertions

References 6,7,8: Repeat, Transmembrane, Circ. permutation

Reference 9: linear motifs

Reference 10: mixed

problem with an alignment : contact [Julie Thompson](#)

problem with the web site : contact [Raymond Ripp](#)

Practical Session: Source of your Sequences



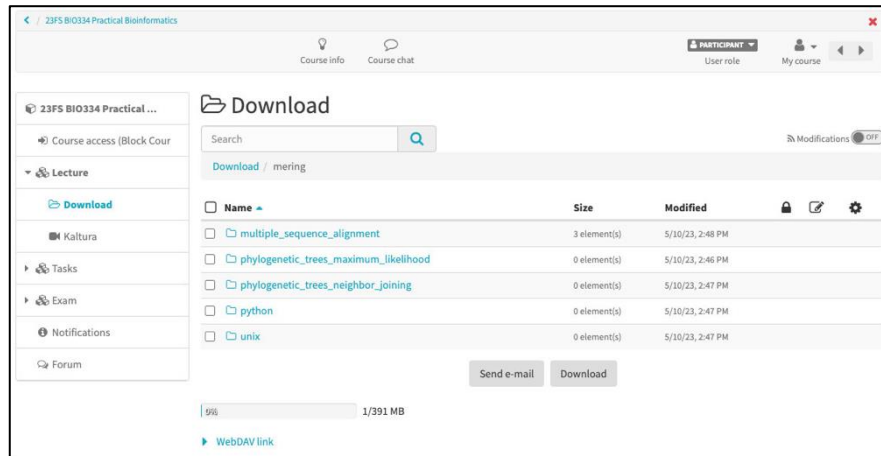
**Dental Calculus of
Medieval Mummy**

(with kind permission of
Christina Warinner, UZH)

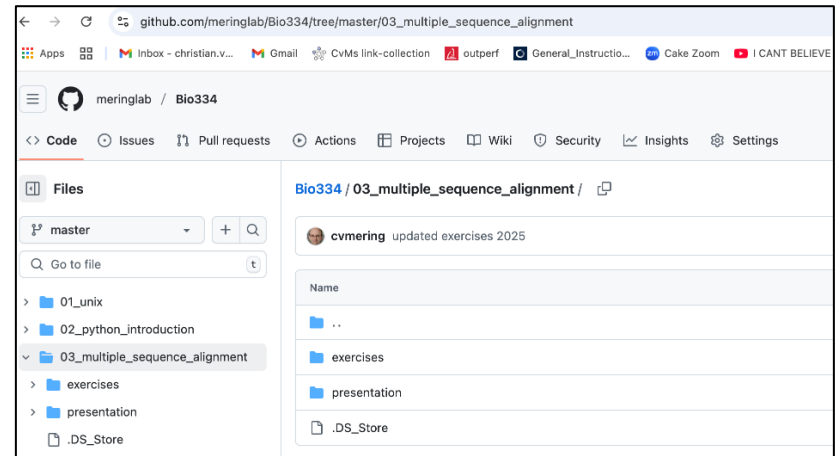
bonus material since 2020:
COVID-19 sequences !

Practical Session: Scripts and Files on OLAT

on OLAT



on github



```
( git clone https://github.com/meringlab/Bio334.git )  
git pull
```

```
Mac: source bio334_env/bin/activate  
Windows: .\bio334_env\Scripts\Activate.ps1
```